

VolSeg Documentation; Installation and Setup Guides

The following guides detail both walkthroughs designed for users with limited coding experience, and developmental users. This documentation covers the setting up of virtual environments, a recommended procedure for installing the Volume-Segmentics package, and alternative installation instructions for more advanced users using *poetry*.

Conda/Virtual Environments

When using Volume Segmentics (Vol-Seg), the recommended pathway to getting set up with an environment is through Anaconda (conda); an open source, user friendly and well-known package management tool. Conda is used primarily due to its versatility and compatibility with other packages most commonly used in conjunction with Vol-Seg (Napari, JupyterLab). Instructions on how to download conda can be found on the [main software page](#), and instructions as to its user functionality can be found on the [main documentation site](#).

Once downloaded, open a terminal instance and create a virtual environment to house your installation of VolSeg. It is also advisable that you create a folder in your user space in an easily accessible place to store your Vol-Seg and other environments you may create in the future; an example would be `/home/'user_space'/envs`. Creating an environment is done through a command line terminal using the following format, alongside deactivating it once you are finished;

1 - Navigate to installation location using the 'cd' command; e.g. `cd /home/'user_space'/'envs`

- This can be done in stages using individual directory names until you reach the required location.
- 2 - Create an environment using the 'conda create' command; `conda create "path_to_env" python="py_version"`
- "path_to_env" refers to the full directory path of the environment; `/home/'user'/envs/"env_name"` etc.

- "env_name" refers to the name of your environment; make the name easy to remember with no spaces e.g. 'VolSeg_env'

- "py_version" refers to the version of Python used within the environment; this should be version 3.10 or higher

2 - Activate the environment using the 'conda activate' command; `conda activate "path_to_env"`

- "path_to_env" refers to the name of your full environment path including the environment name

3 - Deactivate your environment when you are finished using the 'conda deactivate' command and close the terminal; `conda deactivate "path_to_env"`

```
cd "/home/'user_space'/'envs'"
conda create -y -p "path_to_env/" python="py_version" #Python version should be
3.10 or higher
```

```
conda activate "path_to_env"
conda deactivate "path_to_env/"
```

A helpful selection of Python tutorials can be found at [W-3-Schools](#), and further information relating to Python documentation can be found [here](#) if you are not familiar with coding formats.

VolSeg Installation; Default Walkthrough (PyPI)

After an environment has been created, make sure it is activated, then navigate back to your user space (`cd '/home/'user_space'` or similar). The VolSeg Package should be installed in a directory separate from your environment.

It is advisable that you create a folder in your user space an easily accessible space to store the VolSeg Installation, where the repository can be navigated to easily while viewing and altering its scripts; an example would be `/home/'user_space'/libs`, where "libs" refers to a package "library" designation. You can use a 'libs' folder (or equivalent) you currently use, or an alternate location that is memorable; one can be created from the command line using the following steps, or it can be manually created using your file explorer; you can then install the Vol-Seg package using *pip*.

1 - Navigate to your user space within your terminal using the 'cd' command; `cd "/home/'user_space'"`

2 - Make a 'libs' (library) directory; `mkdir "libs"`, or further navigate to an easily accessible directory where you wish to house the Vol-Seg installation using the 'cd' command

- You will need to navigate into the newly made "libs" directory if one is made; `cd "users/'Individual_User'/'libs'"`

3 - Install the Vol-Seg package using *pip*; it should then appear in the 'libs' folder you have created/selected once finished;

```
cd /users/'Individual_User' #Navigate to your user space/desired directory.
mkdir libs #Make a 'libs' directory (if needed).
cd /users/'Individual_User'/libs #Navigate to your housing/'libs' directory.

pip install volume-segmentics
```

Once installed, you will need to navigate into your installed VolSeg directory to begin using its commands; the commands are only executable from within the this directory, relying on the settings within the installation to execute.

```
cd volume-segmentics
```

If a CUDA-enabled build of PyTorch is not being installed by pip, you can use the adaptations below;

```
pip install volume-segmentics --extra-index-url https://download.pytorch.org/whl
```

- If you require different versions of PyTorch/CUDA as per your computer build capabilities (GPU), these can further be modified to suit your needs; uninstall the current version of PyTorch installed within the Vol-Seg installation, and reinstall the correct/preferred version.

```
pip uninstall torch #Uninstall previous torch dependencies.  
pip install torch=='PyTorch_Version' --index-url https://download.'full link'  
#Install new torch dependencies.
```

- Information as to the versions of PyTorch/CUDA available can be found using the python [get started](#) link; the download path should be marked with the argument `--index-url` alongside the versions you require. Previous versions of PyTorch can also be found [here](#).

Opt. To keep track of your segmentation history, you can optionally install '*Tensorboard*'; a package that keeps track of your training and prediction outputs. Further information about using tensorboard can be found [here](#).

```
pip install tensorboard
```

VolSeg Installation; Advanced version (Poetry)

The packages needed to use Vol-Seg and manage releases, updates and amend software and package parameters, can also be installed alternatively through 'poetry'; a dependency and packaging tool. This installation method is suited towards developmental users or those who wish to interact with installations on a deeper level.

To install Vol-Seg in this way, you should first complete the initial steps outlined above; create and activate a virtual environment and create or navigate to a 'libs' folder (or alternative user directory /space) using a terminal. You can then follow the subsequent steps to install VolSeg;

1 - Make sure you have your Vol-Seg conda environment activated and are navigated into your "libs" directory from the terminal; `conda activate "path_to_env/env_name"` and `cd "users/'Individual_User'/'libs'"`

2 - Clone the Volume-Segmentics package directly from the online Git repo using the '*git clone*' command; this will then appear as a 'volume-segmentics' directory within your chosen or equivalent 'libs' folder: `"git clone --branch 'release_version' http://'Git_link'"`

3 - Navigate into the cloned Vol-Seg directory using the `cd` command; `cd "users/'Individual_User'/'libs'/volume-segmentics"`

4 - Install the poetry tool using pip; `pip install poetry`

5 - Install volume-segmentics using poetry; `poetry install`

- The poetry tool will know the packages required for the installation by reading the cloned repo VolSeg file dependencies.
- You may also be required to run the `*'poetry lock'*` command, depending on your computer setup; this will be prompted as an error message when to try to execute poetry install, where the `'poetry install'` command can be rerun after `'poetry lock'` has finished.

```
conda activate "path_to_env/env_name" #Activate virtual environment
cd "users/'Individual_User'/'libs'" #Navigate to suitable user space
git clone --branch vs_'Version' https://github.com/'full_link'.git #Clone the
VolSeg GitHub repo.
cd /ceph/users/'Individual_User'/libs/volume-segmantics #Navigate into the cloned
VolSeg directory.
pip install poetry #Install Poetry
poetry install #Use Poetry to install VolSeg.
```